A First Course in Causal Inference Ch. 11

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Ch. 11 The Central Role of the Propensity Score in Observational Studies for Causal Effects

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- Rosenbaum and Rubin (1983) proposed the key concept *propensity score*.
- Under the i.i.d. sampling assumption, we have four random variables associated with each unit: $\{X, Z, Y(1), Y(0)\}$.
- We can factorize the joint distribution as

$$\Pr\{X, Z, Y(1), Y(0)\} = \Pr\{X\} \cdot \Pr\{Y(1), Y(0) \mid X\} \cdot \Pr\{Z \mid X, Y(1), Y(0)\}.$$

- We do not want to model the covariates becasue they are background information happening before the treatment and outcome.
- If we want to go beyond the outcome model, then we must focus on the treatment assignment mechanism, which leads to the definition of the propensity score.

Definition 11.1 (propensity score)

Define

$$e(X, Y(1), Y(0)) = \Pr\{Z = 1 \mid X, Y(1), Y(0)\}$$

as the propensity score. Under strong ignorability, we have

$$e(X, Y(1), Y(0)) = \Pr\{Z = 1 \mid X, Y(1), Y(0)\} = \Pr(Z = 1 \mid X),$$

so the propensity score reduces to

$$e(X) = \Pr\{Z = 1 \mid X\},$$

the conditional probability of receriving the treatment given the observed covariates.

11.1 The propensity score as a dimension reduction tool

Theorem 11.1

$Z \perp\!\!\!\perp \{Y(1), Y(0)\} \mid X \Rightarrow Z \perp\!\!\!\perp \{Y(1), Y(0) \mid e(X)\}$.

- Thm. 11.1 implies that conditioning on the propensity score $e(X)$ removes all confunding induced by covariates X.
- The original covariates X can be high-dimensional but $e(X)$ is one-dimensional scalar variable bounded between 0 and 1.
- Therefore, the propensity score reduces the dimension of the original covariates but still maintains the ignorability: a *dimensional reduction tool*.

Proof.

We need to show

$$\Pr\{Z = 1 \mid Y(1), Y(0), e(X)\} = \Pr\{Z = 1 \mid e(X)\}.$$

The left-hand side of the equation equals:

$$\begin{aligned}\Pr\{Z = 1 \mid Y(1), Y(0), e(X)\} &= \mathbb{E}[Z \mid Y(1), Y(0), e(X)] \\ &= \mathbb{E}[\mathbb{E}[Z \mid Y(1), Y(0), e(X), X] \mid Y(1), Y(0), e(X)] \quad (\because \text{LIE.}) \\ &= \mathbb{E}[\underbrace{\mathbb{E}[Z \mid X]}_{=e(X)} \mid Y(1), Y(0), e(X)] = e(X).\end{aligned}$$

The right-hand side of the equation equals:

$$\Pr\{Z = 1 \mid e(X)\} = \mathbb{E}[Z \mid e(X)] = \mathbb{E}[\mathbb{E}[Z \mid e(X), X] \mid e(X)] \quad (\because \text{LIE.}) = \mathbb{E}[\underbrace{\mathbb{E}[Z \mid X]}_{=e(X)} \mid e(X)] = e(X)$$

□

11.1 The propensity score as a dimension reduction tool

- Thm. 11.1 motivaes *propensity score stratification*:

$$Z \perp\!\!\!\perp \{Y(1), Y(0)\} \mid e(X) = e_k \quad (k = 1, \ldots, K),$$

  where the propensity score only takes K posible values $\{e_1, \ldots, e_K\}$.
- In general, the propensity score is not known and is not discrete.
- We often fit a statistical model for $\Pr\{Z = 1 \mid X\}$ to obtain $\hat{e}(X)$ and we can discretize it to approximate the simple case above.
- For example, we can discretize $\hat{e}(X)$ by its K quantiles to obtain $\hat{e}'(X)$: $\hat{e}'(X) = e_k$, the k/K-th quantile of $\hat{e}(X)$, if $\hat{e}(X)$ is between the $(k-1)/K$-th and k/K-th quantitles of the $\hat{e}(X)$'s.

11.1 The propensity score as a dimension reduction tool

- Its rigorous quantification is still missing in the literature.
- Q1: How do we choose K?
  1. If K is too small, ignorability does not hold conditional on $\hat{e}'(X)$ even approximately.
  2. If K is too large, then we do not have enough units within each stratum of the estimated propensity score and many strata have only treated or control units.
- Q2: How do we compute standard errors of the estimators based on propensity score stratification?
  1. Reporting standard errors based on the SRE ignores the uncertainty in the estimated propensity scores.
  2. Some researchers employ the bootstrap method but the theory for the bootstrap is still unclear due to the discreteness of this estimator.

11.2 Propensity score weighting

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References I

Rosenbaum, P. R. and D. B. Rubin (1983). The central role of the propensity score in observational studies for causal effects. *Biometrika 70*(1), 41–55.